Introduction

The vector space model is the mathematical representation so that the text document is represented in a high-dimensional vector. This concept is used in both natural language processing and information retrieval systems. A document is a combination of many words. These words are represented by a dimension, so the dimension values show how important, frequent, or rare that phrase is in the document. Then the similarity between the query and the document is measured by calculating similarity scores. Therefore, it is the text mining process for defining the queries and documents so that they are measured in high-dimensional space in that vector space, which is the vector space model. It has been done so that this step is very beneficial for calculating the similarity scores between documents and queries.

Below is the step for how it ranks the document in the vector space model.

Document representation: Every document is the collation of the vector in the vector space model. It has the vocabulary phrases in a vector dimension so that the values of the dimensions indicate the terms that are relevant information in the document. For example, we use TF, which measures the term frequency as commonly used words are included here, and IDF, which is the inverse document frequency. Then these two terms will be mixed to give the measure of the term. TF gives the common word in the document and IDF gives the rare used words in document.

Query Representation: In the query, a collection of words with their related weights is used so that the query also represents the same vector space. There is a term present in the query, so it provides the appropriate value to match it. Its TF-IDF has also been calculated.

Similarity Calculation: When the query and the documents are in the form of vectors in the vector space model, then we use cosine similarity to calculate the similarity scores. The cosine similarity measures the cosine angle between the two vectors. The range between the two vectors is from -1 to 1. The score 1 indicates there is perfect similarity, the score o indicates there is no similarity, and the score -1 indicates there is perfect dissimilarity.

Document Ranking: Then it is ranked as per their similarity scores, so that the similarity scores between the query and the document are calculated in every corpus. If there are higher similarity scores, it shows the document is more related to the given search query, and it appears higher in the result.

Presentation and retrieval: At the end, the top-ranking documents provided by the user are often presented in descending order based on similarity ratings. These documents indicate that the vector space model is more relevant to the user query.

Relevance Threshold: This relevance threshold approach is to filter low-similarity score items. Then the most relevant information is returned in the search result using this concept.